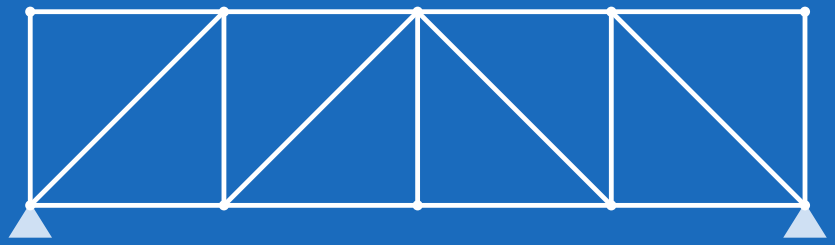
TRUSS REPORT

Howe truss

Plane truss — analysis and CIRSOC 301 rod checks

Author A. Student · Course Structures I · 13/10/2026



MAX TENSION	MAX COMPRESSION	DEFLECTION	MOST USED ROD	BUCKLING Λ_1	WEIGHT
+80.0 kN	**-84.9 kN**	**5.9 mm**	**0.69**	**1.90**	**542 kg**
rod 1	rod 13	L/2020 · limit L/300	rod 13 · inelastic buckling (E3	the load can grow ×1.9	carries 120.0 kN · 23× own weight

The model

Span	12.00 m
Height	3.00 m
Nodes / rods	10 / 17
Supports	pin at 0, rollerX at 4
Loads	3 nodal, total 120.0 kN
Rod families	3, with their sections:
Chord	CHS 114.3x4.5, F24 (CIRSOC) / A36
Diagonal	CHS 101.6x4.0, F24 (CIRSOC) / A36
Vertical	CHS 60.3x3.6, F24 (CIRSOC) / A36

What the analysis found

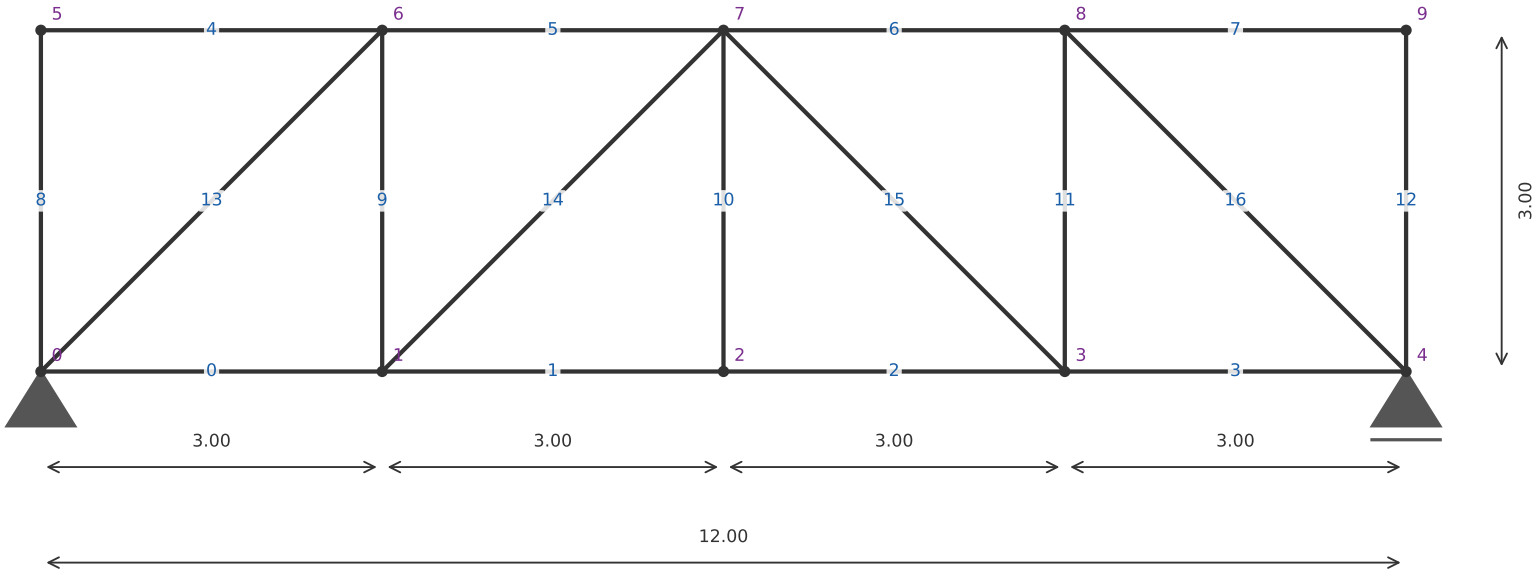
Stability	Statically determinate (m + r − 2j = 17 + 3 − 2×10 = 0): statics alone gives every force.
Most used rod	rod 13 at 0.69 (inelastic buckling (E3-2))
Rods over capacity	0 of 17 checked
Buckling	$\lambda_1 = 1.90$
Shear and moment	none: every rod is pinned and loaded only at its nodes, so the axial force is the whole result

Contents

2 Geometry
3 Axial forces
4 Deflection and stress
5 Reactions
6 Utilisation
7 Buckling
8 Node force vectors · 2 sheets
10 Rods
11 Rod calculations · 2 sheets
13 Explain rod 13 and 16

Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

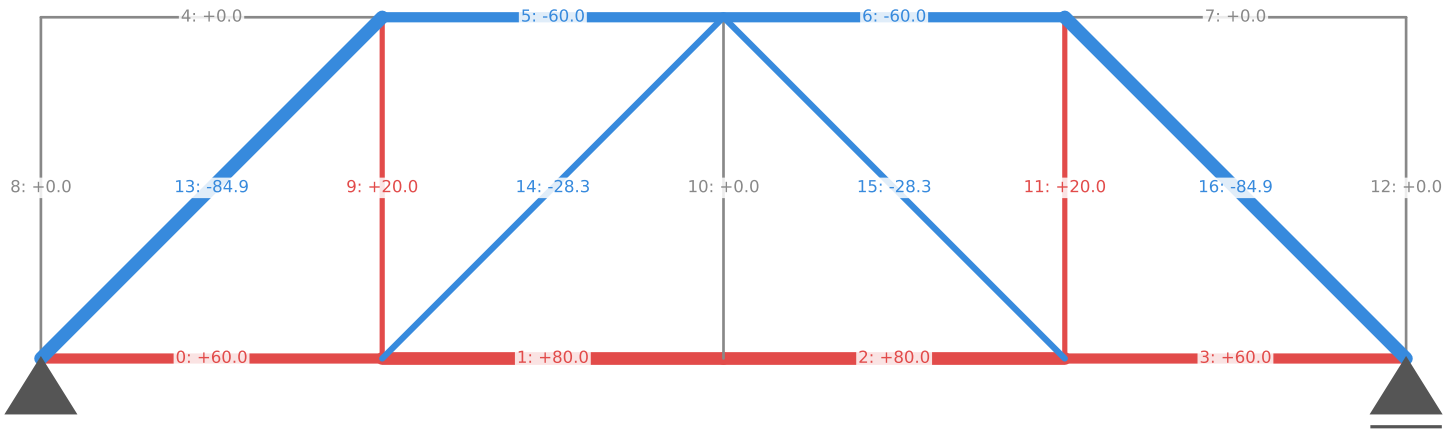
Node numbers in purple, rod numbers in blue. Dimensions in m.



2 m · true to scale

0 2

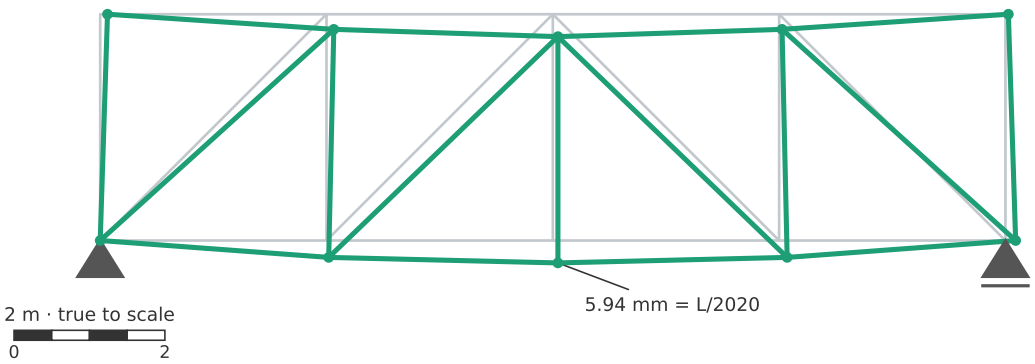
Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 84.9).



2 m · true to scale

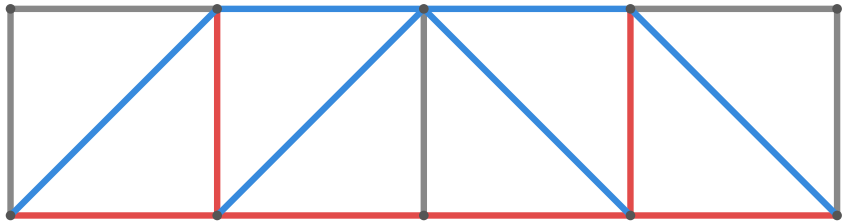
0 2

Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50. Every rod is pinned, so every member stays straight: there is no curvature, and no contraflexure. Largest vertical deflection 5.94 mm against the limit L/300 = 40.0 mm: OK.

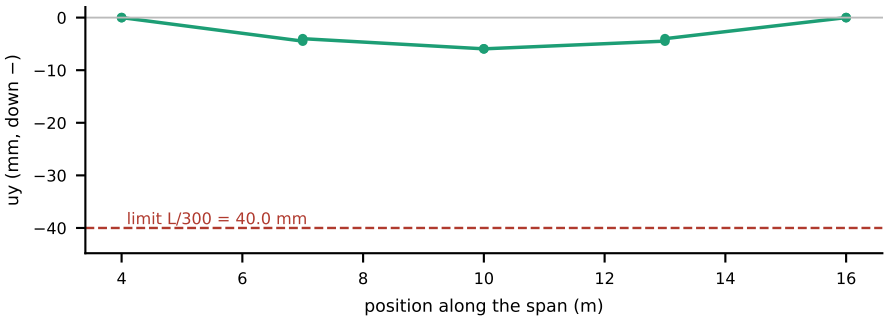


Fibre stress top and bottom face of each member

red = tension, blue = compression. Every rod is pinned, so both faces of a rod carry the same stress and each rod is one colour: the axial force.

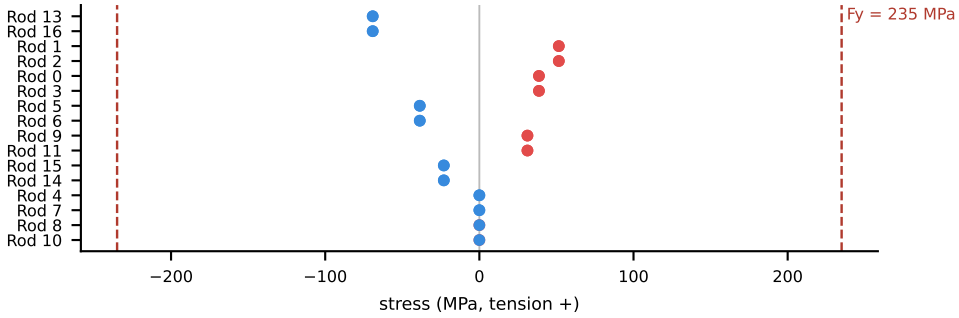


Deflection along the span vertical movement of every joint



Largest 5.94 mm at joint 2 = L/2020. Within the limit.

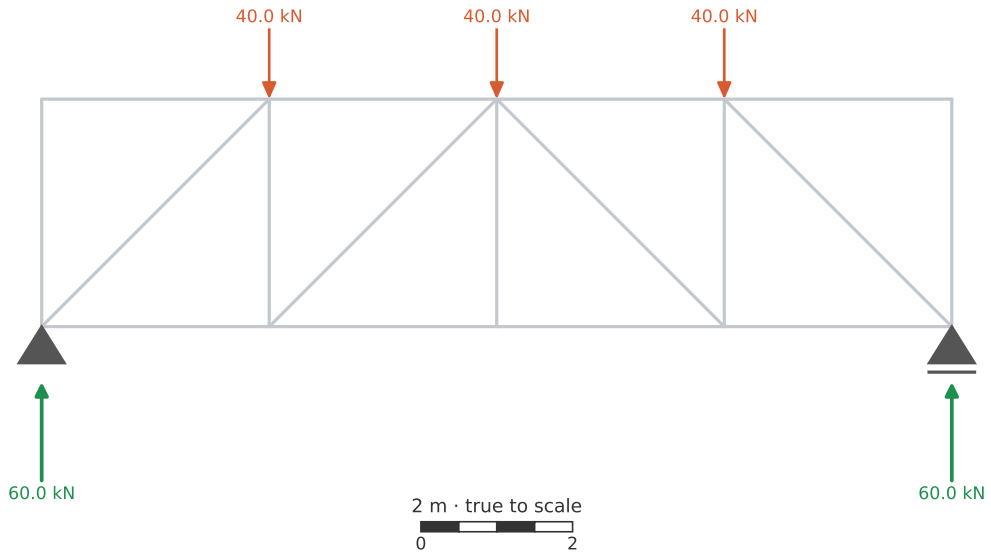
Fibre stress values the two faces of each rod, MPa



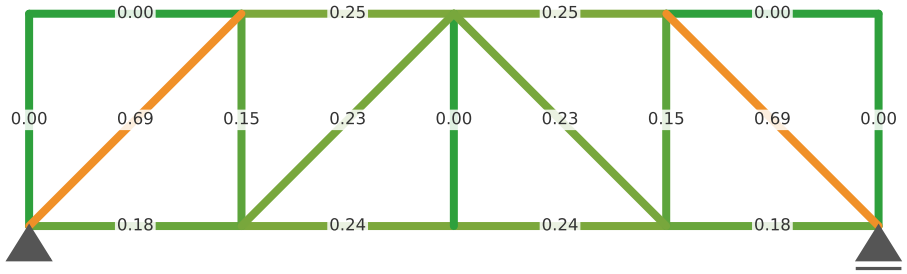
Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

Support reactions (green arrows), as the supports push on the truss, against the loads (orange); Ry positive upwards. The sums balance the loads.

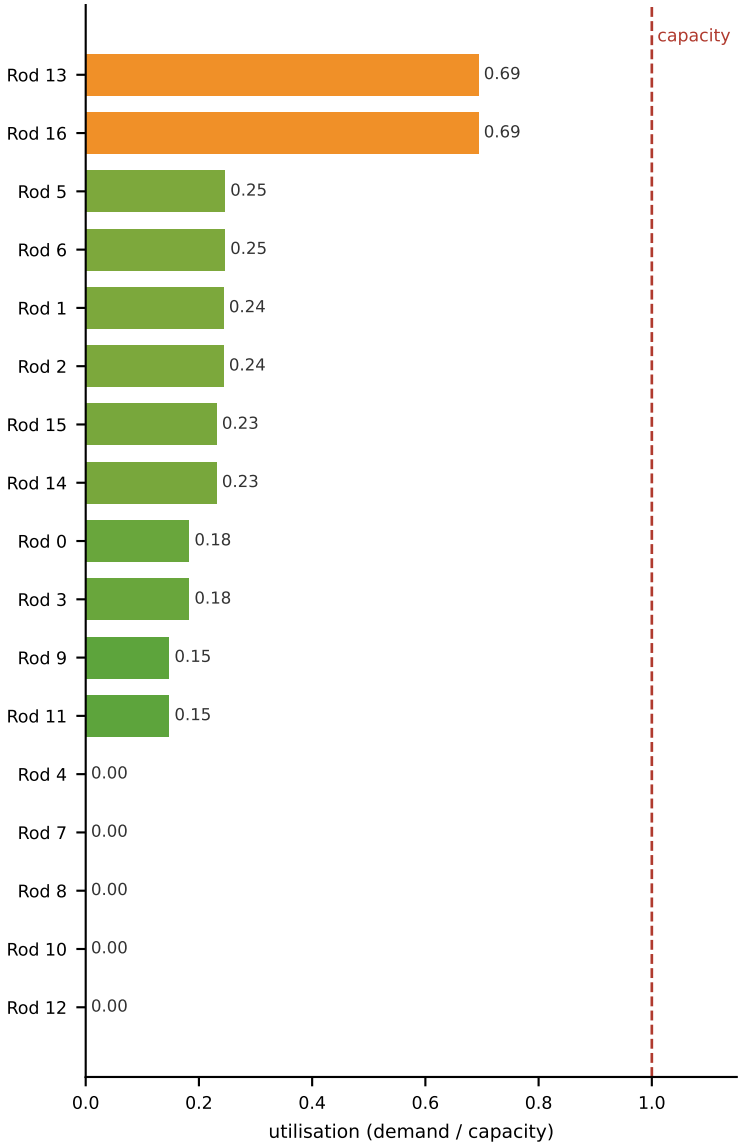
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	60.00
4	rollerX	0.00	60.00
Σ		0.00	120.00

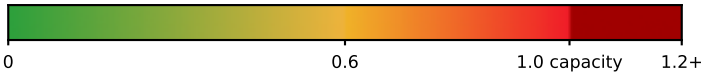


Utilisation: demand over CIRSOC 301 capacity. Most used rod 13 at 0.69 (inelastic buckling (E3-2)). Every checked rod is within capacity.



2 m · true to scale
0 2

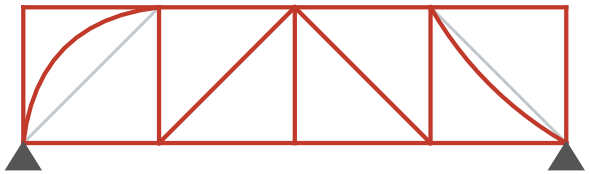




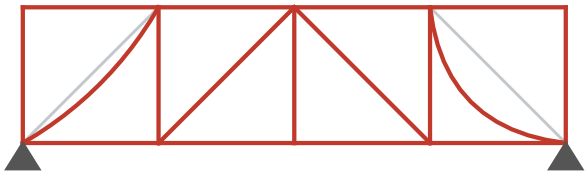
grey = no steel section, not checked

λ is the factor on the load as it stands. $\lambda_1 = 1.90$, so the load can grow 1.9× before it buckles. 2 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

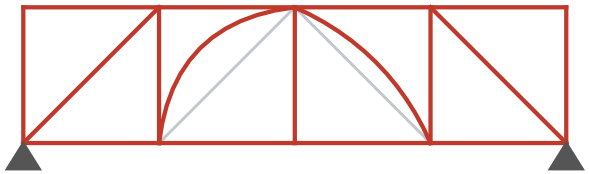
Mode 1: λ = 1.90
rod 13 bows

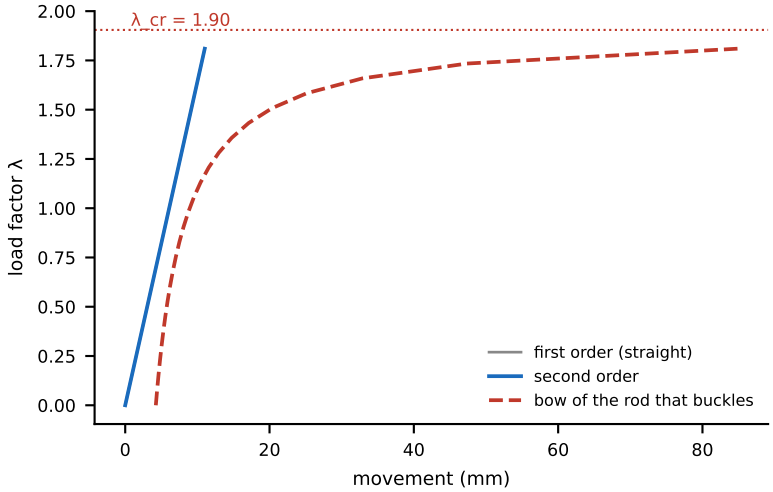


Mode 2: λ = 1.90
rod 16 bows



Mode 3: λ = 5.71
rod 14 bows





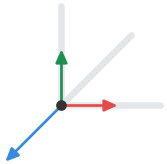
The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 4.24 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 0 · pin

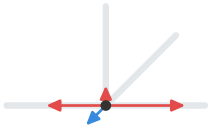




member	N	θ°	Fx	Fy
r0→1	+60.0	0.0	+60.0	+0.0
r8→5	+0.0	180.0	+0.0	+0.0
r13→6	-84.9	225.0	-60.0	-60.0
reaction			+0.0	+60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 1

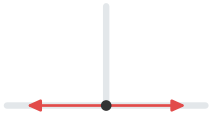




member	N	θ°	Fx	Fy
r0→0	+60.0	180.0	-60.0	+0.0
r1→2	+80.0	0.0	+80.0	+0.0
r9→6	+20.0	90.0	+0.0	+20.0
r14→7	-28.3	225.0	-20.0	-20.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 2

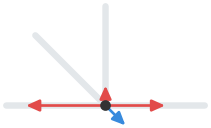




member	N	θ°	Fx	Fy
r1→1	+80.0	180.0	-80.0	+0.0
r2→3	+80.0	0.0	+80.0	+0.0
r10→7	+0.0	180.0	+0.0	+0.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 3

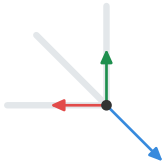




member	N	θ°	Fx	Fy
r2→2	+80.0	180.0	-80.0	+0.0
r3→4	+60.0	0.0	+60.0	+0.0
r11→8	+20.0	90.0	+0.0	+20.0
r15→7	-28.3	315.0	+20.0	-20.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 4 · rollerX





member	N	θ°	Fx	Fy
r3→3	+60.0	180.0	-60.0	+0.0
r12→9	+0.0	180.0	+0.0	+0.0
r16→8	-84.9	315.0	+60.0	-60.0
reaction			+0.0	+60.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 5

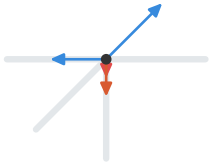




member	N	θ°	Fx	Fy
r4→6	+0.0	180.0	+0.0	+0.0
r8→0	+0.0	180.0	+0.0	+0.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 6

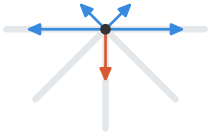




member	N	θ°	Fx	Fy
r4→5	+0.0	0.0	+0.0	+0.0
r5→7	-60.0	180.0	-60.0	+0.0
r9→1	+20.0	270.0	+0.0	-20.0
r13→0	-84.9	45.0	+60.0	+60.0
load			+0.0	-40.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 7



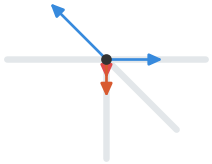


member	N	θ°	Fx	Fy
r5→6	-60.0	0.0	+60.0	+0.0
r6→8	-60.0	180.0	-60.0	+0.0
r10→2	+0.0	180.0	+0.0	+0.0
r14→1	-28.3	45.0	+20.0	+20.0
r15→3	-28.3	135.0	-20.0	+20.0
load			+0.0	-40.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Every force ON each joint, drawn from its centre: each member's pull (red, tension) or push (blue, compression), the load (orange), the reaction (green). x right, y up, θ anticlockwise from +x. Σ ≈ 0: the joint is in equilibrium; these are the forces its gusset plate is designed for.

Joint 8





member	N	θ°	Fx	Fy
r6→7	-60.0	0.0	+60.0	+0.0
r7→9	+0.0	180.0	+0.0	+0.0
r11→3	+20.0	270.0	+0.0	-20.0
r16→4	-84.9	135.0	-60.0	+60.0
load			+0.0	-40.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Joint 9





member	N	θ°	Fx	Fy
r7→8	+0.0	0.0	+0.0	+0.0
r12→4	+0.0	180.0	+0.0	+0.0
Σ kN	Fx 0.00	Fy 0.00	✓	

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
0	0–1	Chord	CHS 114.3x4.5	3.00	+60.00	normal stress (H.3.4)	0.18 OK
1	1–2	Chord	CHS 114.3x4.5	3.00	+80.00	normal stress (H.3.4)	0.24 OK
2	2–3	Chord	CHS 114.3x4.5	3.00	+80.00	normal stress (H.3.4)	0.24 OK
3	3–4	Chord	CHS 114.3x4.5	3.00	+60.00	normal stress (H.3.4)	0.18 OK
4	5–6	Chord	CHS 114.3x4.5	3.00	+0.00	inelastic buckling (E3-2)	0.00 OK
5	6–7	Chord	CHS 114.3x4.5	3.00	-60.00	inelastic buckling (E3-2)	0.25 OK
6	7–8	Chord	CHS 114.3x4.5	3.00	-60.00	inelastic buckling (E3-2)	0.25 OK
7	8–9	Chord	CHS 114.3x4.5	3.00	+0.00	inelastic buckling (E3-2)	0.00 OK
8	0–5	Vertical	CHS 60.3x3.6	3.00	+0.00	normal stress (H.3.4)	0.00 OK
9	1–6	Vertical	CHS 60.3x3.6	3.00	+20.00	normal stress (H.3.4)	0.15 OK
10	2–7	Vertical	CHS 60.3x3.6	3.00	+0.00	normal stress (H.3.4)	0.00 OK
11	3–8	Vertical	CHS 60.3x3.6	3.00	+20.00	normal stress (H.3.4)	0.15 OK
12	4–9	Vertical	CHS 60.3x3.6	3.00	+0.00	normal stress (H.3.4)	0.00 OK
13	0–6	Diagonal	CHS 101.6x4.0	4.24	-84.85	inelastic buckling (E3-2)	0.69 OK
14	1–7	Diagonal	CHS 101.6x4.0	4.24	-28.28	inelastic buckling (E3-2)	0.23 OK
15	3–7	Diagonal	CHS 101.6x4.0	4.24	-28.28	inelastic buckling (E3-2)	0.23 OK
16	4–8	Diagonal	CHS 101.6x4.0	4.24	-84.85	inelastic buckling (E3-2)	0.69 OK

Rod calculations (1/2)

Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

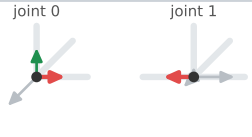
Rod 0

joint 0 → 1

tension



```
L = 3.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +60.00 kN
at 0:  Fx = +60.00   Fy = +0.00
at 1:  Fx = -60.00   Fy = +0.00
```



Rod 1

joint 1 → 2

tension



```
L = 3.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +80.00 kN
at 1:  Fx = +80.00   Fy = +0.00
at 2:  Fx = -80.00   Fy = +0.00
```



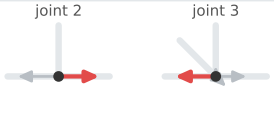
Rod 2

joint 2 → 3

tension



```
L = 3.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +80.00 kN
at 2:  Fx = +80.00   Fy = +0.00
at 3:  Fx = -80.00   Fy = +0.00
```



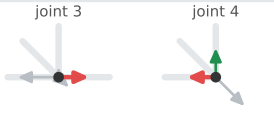
Rod 3

joint 3 → 4

tension



```
L = 3.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +60.00 kN
at 3:  Fx = +60.00   Fy = +0.00
at 4:  Fx = -60.00   Fy = +0.00
```



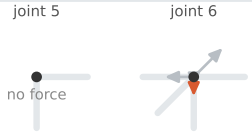
Rod 4

joint 5 → 6

no force



```
L = 3.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +0.00 kN
at 5:  Fx = +0.00   Fy = +0.00
at 6:  Fx = +0.00   Fy = +0.00
```



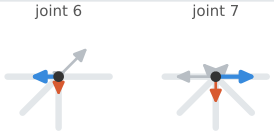
Rod 5

joint 6 → 7

compression



```
L = 3.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = -60.00 kN
at 6:  Fx = -60.00   Fy = +0.00
at 7:  Fx = +60.00   Fy = +0.00
```



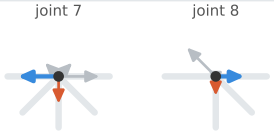
Rod 6

joint 7 → 8

compression



```
L = 3.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = -60.00 kN
at 7:  Fx = -60.00   Fy = +0.00
at 8:  Fx = +60.00   Fy = +0.00
```



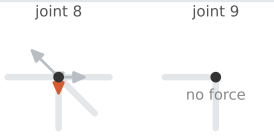
Rod 7

joint 8 → 9

no force



```
L = 3.000 m     θ = 0.0°
cos θ = +1.000   sin θ = +0.000
N = +0.00 kN
at 8:  Fx = +0.00   Fy = +0.00
at 9:  Fx = +0.00   Fy = +0.00
```



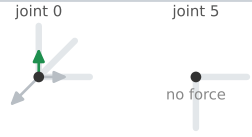
Rod 8

joint 0 → 5

no force



```
L = 3.000 m     θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +0.00 kN
at 0:  Fx = +0.00   Fy = +0.00
at 5:  Fx = +0.00   Fy = +0.00
```



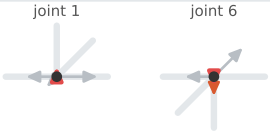
Rod 9

joint 1 → 6

tension



```
L = 3.000 m     θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +20.00 kN
at 1:  Fx = +0.00   Fy = +20.00
at 6:  Fx = +0.00   Fy = -20.00
```



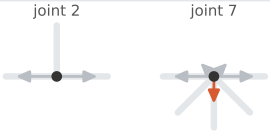
Rod 10

joint 2 → 7

no force



```
L = 3.000 m     θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +0.00 kN
at 2:  Fx = +0.00   Fy = +0.00
at 7:  Fx = +0.00   Fy = +0.00
```



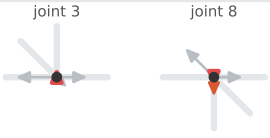
Rod 11

joint 3 → 8

tension



```
L = 3.000 m     θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +20.00 kN
at 3:  Fx = +0.00   Fy = +20.00
at 8:  Fx = +0.00   Fy = -20.00
```



Rod 12

joint 4 → 9

no force



```
L = 3.000 m     θ = 90.0°
cos θ = +0.000   sin θ = +1.000
N = +0.00 kN
at 4:  Fx = +0.00   Fy = +0.00
at 9:  Fx = +0.00   Fy = +0.00
```



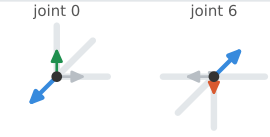
Rod 13

joint 0 → 6

compression



```
L = 4.243 m     θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -84.85 kN
at 0:  Fx = -60.00   Fy = -60.00
at 6:  Fx = +60.00   Fy = +60.00
```



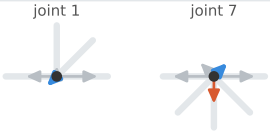
Rod 14

joint 1 → 7

compression



```
L = 4.243 m     θ = 45.0°
cos θ = +0.707   sin θ = +0.707
N = -28.28 kN
at 1:  Fx = -20.00   Fy = -20.00
at 7:  Fx = +20.00   Fy = +20.00
```



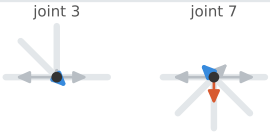
Rod 15

joint 3 → 7

compression



```
L = 4.243 m     θ = 135.0°
cos θ = -0.707   sin θ = +0.707
N = -28.28 kN
at 3:  Fx = +20.00   Fy = -20.00
at 7:  Fx = -20.00   Fy = +20.00
```



Every rod: its length L and angle θ (anticlockwise from +x, from its first joint to its second), its axial force N (+ pulls, − pushes), and what that force does at each end: Fx = N cos θ, Fy = N sin θ on the first joint, and the same reversed on the second. Right: the free body of each end joint, with this rod's force coloured and the rest grey.

Rod 16

joint 4 → 8

compression



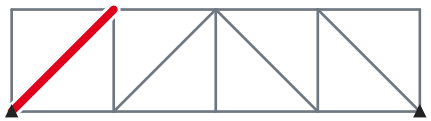
L = 4.243 m θ = 135.0°
cos θ = -0.707 sin θ = +0.707
N = -84.85 kN
at 4: Fx = +60.00 Fy = -60.00
at 8: Fx = -60.00 Fy = +60.00

joint 4



joint 8





Rod 13

utilisation 0.69

inelastic buckling (E3-2)

```
Rod 13
======
Steel, CIRSOC 301-2018 -- A = 1,226 mm², Fy = 235 MPa, L = 4.243 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,243 mm / 34.5 mm
      = 122.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 122.8²
      = 130.80 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 122.8 vs 4.71 √(200,000/235) = 137.4
      = inelastic

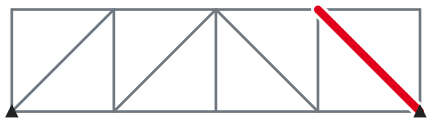
 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/130.80) × 235
      = 110.78 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 110.78 × 1,226
      = 122.29 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 84.85 / 122.29
      = 0.694

 7. Governs
      the largest ratio
      = inelastic buckling (E3-2)
      = 0.694

utilisation 0.694 -- OK
```



Rod 16

utilisation 0.69

inelastic buckling (E3-2)

```
Rod 16
======
Steel, CIRSOC 301-2018 -- A = 1,226 mm², Fy = 235 MPa, L = 4.243 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,243 mm / 34.5 mm
      = 122.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 122.8²
      = 130.80 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 122.8 vs 4.71 √(200,000/235) = 137.4
      = inelastic

 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/130.80) × 235
      = 110.78 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 110.78 × 1,226
      = 122.29 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 84.85 / 122.29
      = 0.694

 7. Governs
      the largest ratio
      = inelastic buckling (E3-2)
      = 0.694

utilisation 0.694 -- OK
```